

AI3403: Multi-Agent Systems (MAS) - Assignment #2

Rules for Submitting Assignment

- The assignment is worth 100 points. You must present the entire work leading to their solution and also attach the associated code.
 - The assignment will be due by **18:00 on Friday, September 04, 2026** in the Google classroom. The deadline is strict and no extensions shall be awarded unless otherwise for unavoidable circumstances.
 - You should submit your assignment as **one** pdf document and the file should be named using roll number. For instance, if your roll number is **ai25btech11111**, then the pdf should be named as **ai25btech11111Assignment1.pdf**. Please also write your name in the first page of your assignment.
 - Students are strongly encouraged to submit their work in a neatly typed $\text{\LaTeX}$ document with all supporting code attached to the same document. An example overleaf latex template can be found in this link: <https://www.overleaf.com/latex/templates/assignment-homework-solutions-template/nfjybpqbdrn>. If typing is not possible, hand-written submissions will be considered as long as they are legibly written and scanned properly with all the codes attached into one pdf.
 - Students can use any coding language of their preference such as Matlab, Python, Julia, C++, etc. All the associated codes for the assignment problems should be attached at the end of each respective problem in your assignment submission. You can also use Jupyter notebook, Google colab, etc to do your assignment and submit it as one pdf.
 - You can use internet, any books, lecture notes, etc. Discussing with other students is perfectly fine and in fact is strongly encouraged. But plagiarism (copying solutions from others without putting your hard work) will never be tolerated.
 - Usage of AI tools like ChatGPT/Claude/Gemini is permitted and must be limited only to understand the concepts taught in the lecture. If you use them in the assignment, you should explicitly declare where and to what extent you have used it. You must never ask AI Tools to solve the entire assignment problems for you. My aim is to educate you and not the AI tools!
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Problem 1: Consider a network of $N \in \mathbb{N}$ agents and the interaction between them is modelled via a graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$. Let the corresponding row-stochastic adjacency matrix be denoted by $A \in \mathbb{R}^{N \times N}$. Let $v \in \mathbb{R}_{\geq 0}^N$ be the normalised left dominant eigenvector such that $v^\top A = v^\top$ and $\mathbf{1}_N^\top v = 1$. For the consensus system $x[k+1] = Ax[k]$, the disagreement vector $\delta[k] \in \mathbb{R}^N$ at time $k \in \mathbb{N}$ is

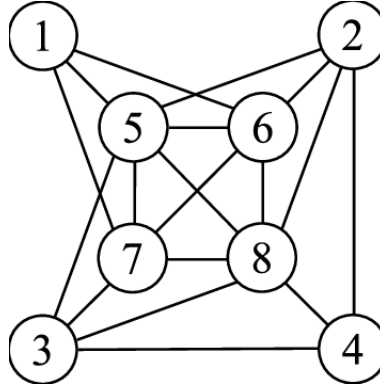
$$\delta[k] = x[k] - (v^\top x[k])\mathbf{1}_N. \quad (1)$$

Note that $v^\top x[k]$ corresponds to the normalized weighted average of the entries of $x[k]$ at time k .

1. Prove that $v^\top x[k]$ is constant over all time steps $k \in \mathbb{N}$.
2. Show that $(A - \mathbf{1}_N v^\top) = (A - \mathbf{1}_N v^\top)(I_N - \mathbf{1}_N v^\top)$
3. Show that the dynamics of the disagreement vector is given by

$$\delta[k+1] = (A - \mathbf{1}_N v^\top)\delta[k]. \quad (2)$$

Problem 2: Simulate the following tasks and give your inferences and all the associated codes.



1. Consider the above 3-robust graph. This offers robustness against $F = 1$ local adversaries in the network. You can assume local adversaries such that each node is allowed to have at most one bad friend. Assume random initial conditions and the adversary is free to do anything. Simulate the resilient consensus of this graph topology using W-MSR protocol.
2. Argue by constructing/drawing subsets to show why this 3-robust graph is *not* a $(3, 2)$ -robust graph. *Hint:* You might need to choose just that one partition S_1, S_2 to prove.
3. Suppose that there is an eavesdropper (aka Spy), who is listening and has access to all the communication links. Note that the eavesdropper is an external entity of the network - who can be either good or bad. For instance, cyber-police are technically “good” eavesdroppers

as they can access our phones for national security purposes. Consider the case of Good eavesdropper who observes the communication, figures out who the malicious agent is and intervenes by secretly modifying the malicious agent's communicated values to help the benign neighbors. How can then the malicious agent execute his attack? What could be the potential strategy of the malicious agent to get past the eavesdropper?

4. Consider the setting where benign agents somehow sense that there is some potential uncontrollable adversary in the network. So, they decide to implement a secure communication, where every node pair (i,j) in the network secretly implements a corresponding encrypted communication. That is, the node pair (i,j) have their own corresponding encryption/decryption setting. This is when you need the combination of both a resilient and a privacy-preserving consensus algorithm for multi-agent systems. Recall that in W-MSR, you do filtering by sorting the neighboring values and then discard the extreme F values and average the remaining ones. Should the benign agents perform W-MSR on the received encrypted neighbor values or should it decrypt the neighbor values first and then perform W-MSR. Justify which one of the above two options is a promising one than the other. What subsequent problems one could face with the promising strategy? There are already papers in this area. Find out and read 2 seminal papers from the research literature to solve this problem and report your understanding (crisply in just 1 paragraph for each paper) of how the authors of the papers addressed this problem. You do not need to code anything here (unless you have all the time on this earth for you!). Remember, this is not a Cryptography class - so no need to report on what cryptographic algorithm was used in those papers. You can happily do that banter in Cryptography courses offered by the CS department.
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